

Testing - Student Course Registration System

In this project, Student Course Registration System, both testing methods including unit and integration testing were implemented to ensure correctness and reliability of the system. The framework that was used for testing was a Python built-in framework named *unittest* along with Flask's test client.

The testing of this system divided into three main parts including unit testing, integration testing, and debugging.

1. Unit testing: the testing was used to test different routes and ensured that each route respond correctly in the Flask application with no dependency on other system components. The tested routes including:
 - Home page redirection
 - Login page
 - Register page access
 - Course, student, enrollment pages
 - Invalid login attempts
2. Integration testing: this testing was used to verify the interaction between multiple components like route access and authentication and made ensure that different part of the system work properly. In addition, session behavior, system stability, and login flow were tested.
3. Debugging: debugging involved using Flask's debugging mode, altering tests not to depend on real data in terms of user interaction, and test output errors such as URL mismatches and session-handling discrepancies.

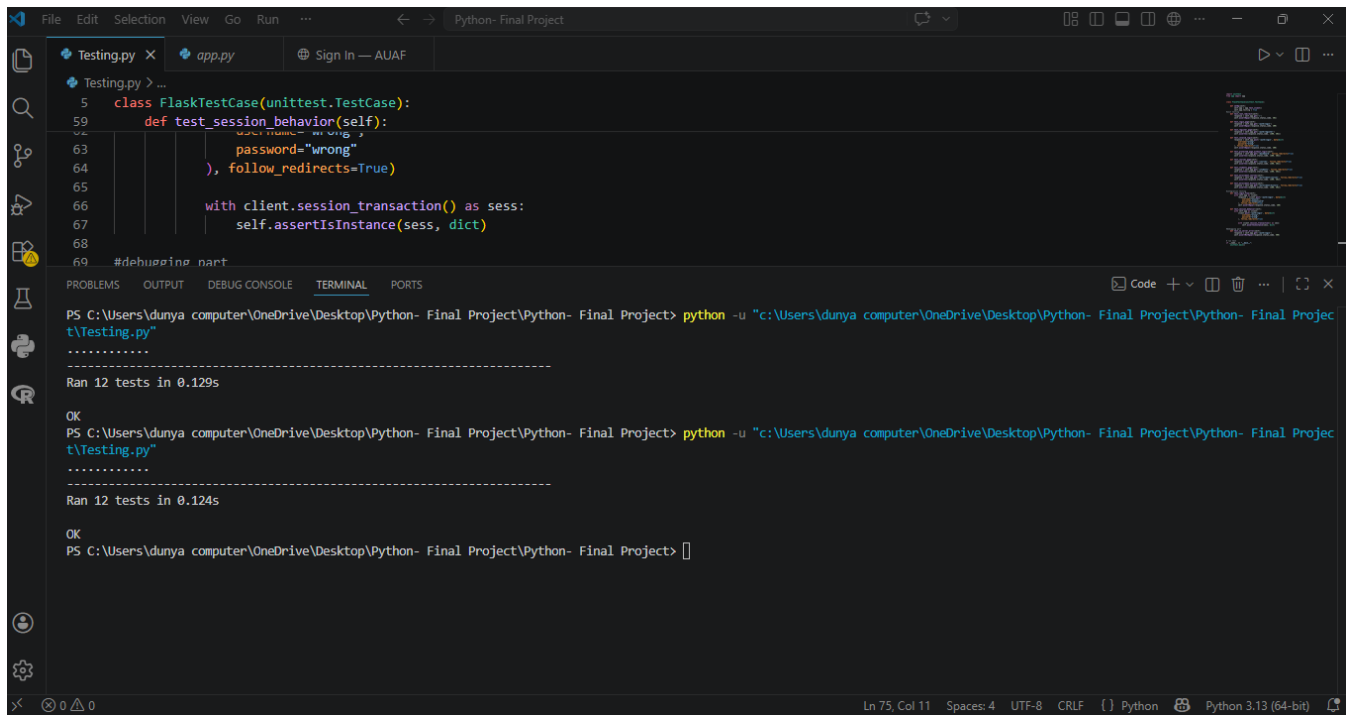
In conclusion, the tests aim at checking the system's response without depending on real data under various conditions. Below is the output of testing indicating that the tests passed successfully.

Final output after testing:

.....

Ran 12 tests in 0.124s

OK



The screenshot shows a Visual Studio Code editor window titled "Python- Final Project". The editor has two tabs: "Testing.py" and "app.py". The "Testing.py" tab is active, showing a Python class `FlaskTestCase` that inherits from `unittest.TestCase`. The class has a method `test_session_behavior` which uses `client.session_transaction()` to interact with a session and `self.assertIsInstance` to verify the session data. The terminal at the bottom shows the command `python -u "c:\Users\dunya computer\OneDrive\Desktop\Python- Final Project\Python- Final Project\Testing.py"` being executed twice. Both executions result in "Ran 12 tests in 0.129s" and "OK".

```
5 class FlaskTestCase(unittest.TestCase):
59     def test_session_behavior(self):
62         #correct wrong
63         password="wrong"
64         ), follow_redirects=True)
65
66         with client.session_transaction() as sess:
67             self.assertIsInstance(sess, dict)
68
69 #dehugging part
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\dunya computer\OneDrive\Desktop\Python- Final Project\Python- Final Project> python -u "c:\Users\dunya computer\OneDrive\Desktop\Python- Final Project\Python- Final Project\Testing.py"
.....
Ran 12 tests in 0.129s

OK
PS C:\Users\dunya computer\OneDrive\Desktop\Python- Final Project\Python- Final Project> python -u "c:\Users\dunya computer\OneDrive\Desktop\Python- Final Project\Python- Final Project\Testing.py"
.....
Ran 12 tests in 0.124s

OK
PS C:\Users\dunya computer\OneDrive\Desktop\Python- Final Project\Python- Final Project>
```

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